MEP report draft

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December 11, 2025

Abstract

For the “real” abstract, please see the README in the GitHub repository.

Thank you for opening this document to learn more about the project. This report is a work-in-progress draft that introduces the key research underpinning the work, as well as the core algorithm and theoretical framework implemented in our model.

Several components are not yet included in this version. These include background on single-molecule chromatin organization (for example, SMC proteins and their interactions with CTCF sites), interpretation of the SCFT results, and, most importantly, a critical assessment of the project’s limitations, particularly those arising from applying mean-field theory to biological systems. This draft may also contain phrasing and grammatical issues; I apologize for them here in advance.

1 Introduction

1.1 Topologically Associated Domain

Eukaryotic genomes are organised in topologically associating domains (TADs). TADs are sub-megabase-scale chromatin compartments within which DNA sequences interact with each other more frequently. TAD boundaries are associated with proteins that regulate the chromosome, including CCCTC-binding factor (CTCF), structural maintenance of chromosome (SMC) such. These boundaries constrain interactions between enhancers and their target genes and therefore play an important role in gene expression.

1.2 Single molecule stuff

SMCs and CTCF

energy level of loop extrusion

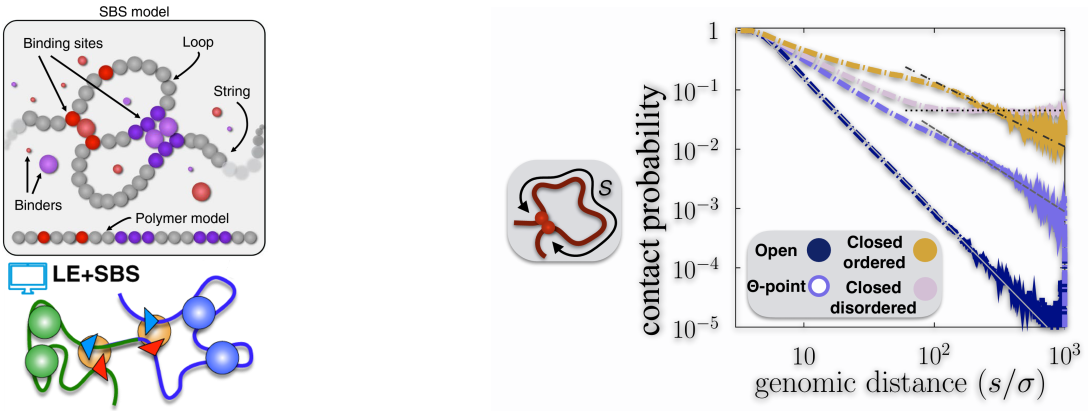
bps per ATP

force measurement from optical tweezer

1.3 Spring-Bead Model

The strings and binders switch (SBS) polymer model [1][2] is a well-established molecular dynamics (MD) simulation method to study the chromosomal 3D organization from Hi-C and FIDH data. The SBS model describes the chromatin filament as a coarse-grained self-avoiding string of beads, which interacts with their cognate molecular binders. Each bead has its ”color” assigned, which corresponds to the property of the DNA sequence, and if two beads have the same color, the free binders in the environment have the opportunity to bind these two beads. The SBS model can be combined with the loop-extrusion (LE) model [3][4], where the molecular binders, in this case an analogy to extruding motor proteins, bind to anchor sites (e.g., CTCF binding sites) on the DNA polymer and dynamically extrude loops.

The SBS model shows that the chromatin polymer can be in different conformations and phases [1]. When the binder concentration and binding energy potential are low, the chromatin is in coil conformation (open), and when they are high, the chromatin is in globule conformation. The contact probability $P_c(s)$ of bead pairs separated by a given contour distance s of different conformations was characterized, as shown in figure 1b. In the coil state, $P_c(s)$ decreases as a power law $P_c(s) \propto s^{-a}$, with an exponent $a \approx 2.1$. At the Θ-point, where the switching between two states happens, the exponent $a \approx 1.5$. In the globule state, $P_c(s)$ first decrease with an exponent close to $a \approx 1.0$, then reach a long plateau.



(a) In the SBS model, chromatin is modeled as a chain of beads. The beads with the same color can be bound by binders in the solvent [?]. The LE+SBS model adds active polymer extrude motors that interact with other motors and CTCF sites.[4]

(b) The power law of contact probability $P_c(s)$ of different polymer conformation states. [1]

Figure 1

The method that assigns "color" to beads is called PRISMR [1][2][5]. It has been shown to have the capability to analyze higher-order chromatin structure from genome-wide Hi-C data. It has the potential to predict the structural variants (SV) result, which can be associated with gene misexpression. Statistical analysis shows that the bead color sequences from PRISMR have a strong correlation with the CTCF binding sites, but they also represent additional interactions of the DNA.

The PRISMR assigns color to beads via the Monte-Carlo method: at each step, a bead is randomly given a color to form a new polymer configuration, and the contact matrix is built based on the new configuration. The difference between this contact matrix and the experimental input, combined with an entropic penalty to prevent overfitting, determines the overall score for the Monte-Carlo acceptance or rejection. The details of this algorithm will be discussed in section !!! where we build our own code to recreate the features of PRISMR, because the original PRISMR code is not publicly available

In the MD simulation of the SBS model, a bead represents a DNA segment of around 30 kb. They are connected via FENE bonds (finitely extensible nonlinear spring) to form the backbone. Weeks-Chandler-Anderson (WCA) potential is used for the repulsive interaction to make the polymer self-avoid. Binders interact with the beads via the Lennard-Jones potential, and loop extrusion motors bind to the polymer as harmonic springs. The Langevin thermostat is used to set the temperature. Each time simulations run up to 10^8 iterations before they reach equilibrium. An ensemble of 10^3 polymer configurations is calculated from parallel simulations for computing the contact matrix, median-distance map, and other properties that characterize the polymer.

The simulation results are then compared with FISH data, where the median radius of gyration R_g gets fit to the microscopy value, to convert the natural unit in the simulation to the real-world scale. The SBS model (and LE+SBS) gives comparable results in the contact map and median distance map (figure 4 b).

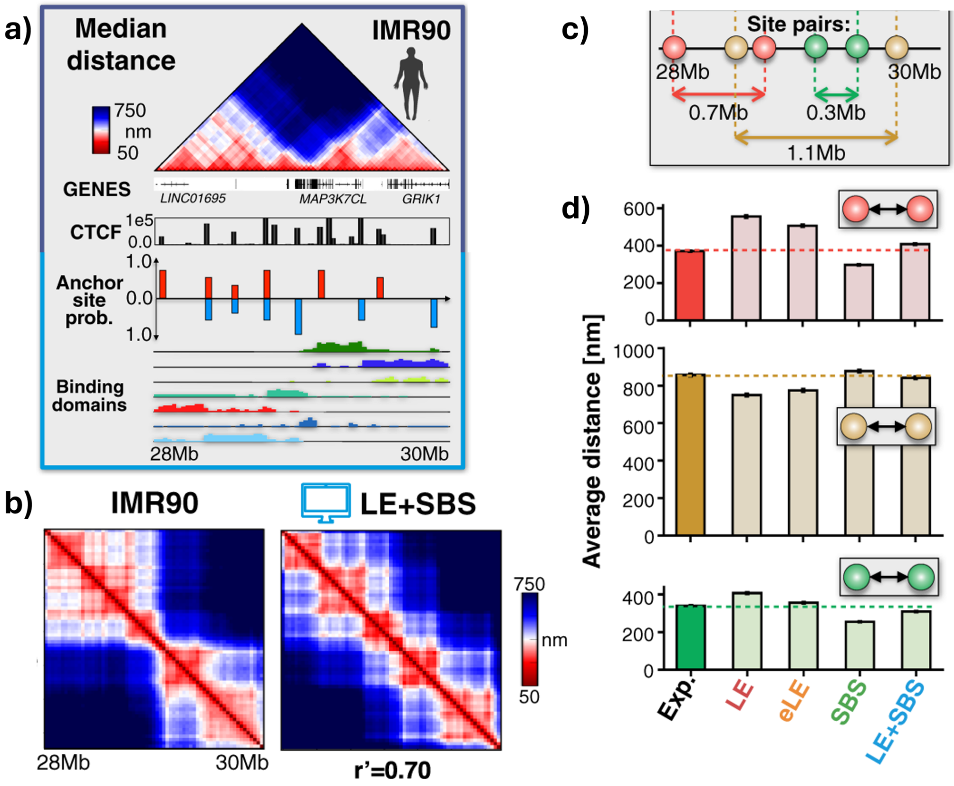


Figure 2: **(a)** Microscopy median distance map and ENCODE CTCF data used in LE model and beads "color" sequence from PRISMR used in SBS model. **(b)** The comparison between microscopy median distance and LE+SBS simulation result. They have a Pearson correlation of $r' = 0.70$. **(c)** and **(d)** The comparison of the average distance of a specific pairs of sites from different models and experiments. [2]

1.4 Other modeling methods

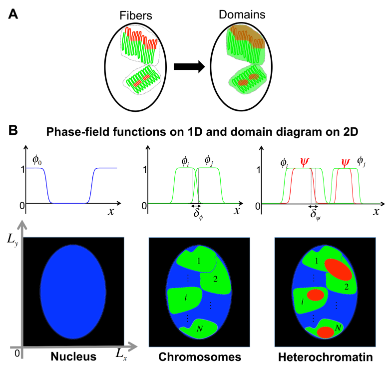


Figure 3: [6]

Pure field-based model with a field formulated resembling to Landau-Ginzburg theory [6].

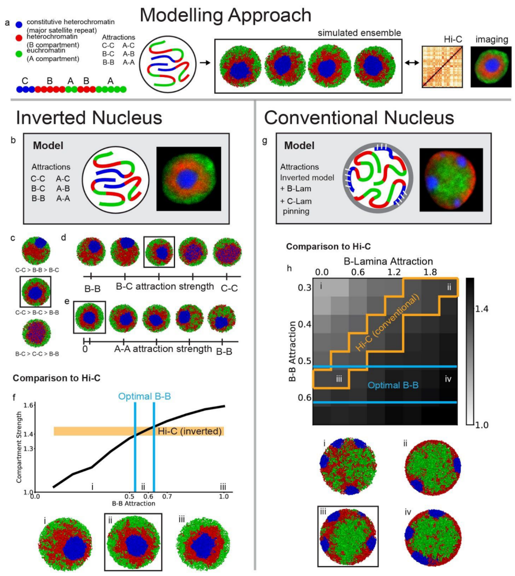


Figure 4: [7]

Another model developed by Mirny's lab [7] uses a coarse-grained polymer beads model with three types of beads to study the inverted nucleus behavior.

1.5 Polymer models

Polymers are long chain-like molecules assembled from small molecules, so-called monomers. The polymerization process, where monomers react and couple into chains, can either happen in biological processes or artificial chemical synthesis. The well-known biological polymers are DNA, polypeptides, polysaccharides, and linear protein assemblies like microtubules. Each of these polymers has vastly different functionalities and physical properties that are crucial to the biological process. Molecular dynamics simulation can be used to study biopolymer-related features at the single-molecule level, such as DNA replication or loop extrusion at full-atom resolution or coarse-grained. However, it is impossible to perform this type of simulation on a large scale due to the computational resource it require. Therefore, polymer models were developed to simplify the interaction while preserving the essential physical properties.[8]

Length scales can be found above which the exact interaction between molecules does not play an important role, and the properties can be described by a few parameters, including persistence length, contour length, charge density, and interaction energy with other molecules.

Some polymers are considered to be very flexible, whose monomers can rotate relatively freely about the carbon-carbon single bonds backbones. Some polymers, such as DNA, which has a double helix backbone, are much more rigid. Those rigid polymers tend to keep their direction over a large distance and are considered semi-flexible or rigid-rod polymers.

freely-joint chain

bead-spring model

continuous gaussian chain

worm-like chain

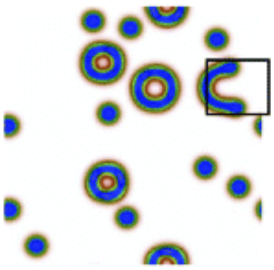
1.6 Self-consistent field theory

Field-based computer simulation is an alternative strategy to the particle-based MD simulation. It utilizes the statistical field theory model, which describes the polymer as continuous fields that vary with position. Rather than the definitive position and velocity of particles, the field-based method includes fields representing the densities of the polymers, and conjugated chemical potential fields that encode the pairwise interactions between polymer elements. [9][8][10]

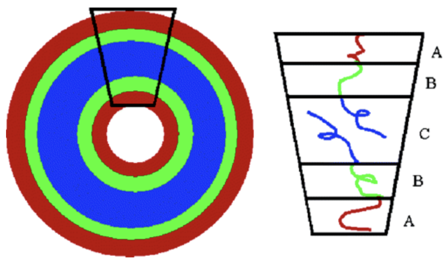
The self-consistent field theory is a powerful theoretical method for studying inhomogeneous polymer systems at equilibrium, firstly introduced by Edwards in the 1960s [11]. One of the most significant applications, by Helfand in 1975 [12], is to predict complex ordered nanodomains or microphases formed by block copolymers, with a relatively simple description of Gaussian chains with short-range contact interactions via Flory-Huggins theory. The details and formulations of SCFT are introduced in the section !!!. [13]

Here we show an example of SCFT's application on block copolymers' self-assembly behavior demonstrated by Rong Wang [14]. A triblock ABC polymer has a first 10% contour length block A being hydrophilic with $\chi_{AS} = 0.5$, the middle 15% block B and the last 75% length block C are hydrophobic, with χ_{BS} and χ_{BS} respectively ranging from 20 to 60. The χ between blocks ranges from 15 to 35. The detailed explanation about the χ parameter is in section !!!. This copolymer phase separates from the solution, and self-assembles into vesicles whose morphology depends on the χ parameters.

Figure 5a shows circular vesicles formed by the copolymers, and figure 5b demonstrates the conformation of the polymers of how they form the vesicles.



(a) The ABC tri-block copolymers self-assemble into vesicles. [14]



(b) The scheme of polymer conformation represented in the mean-field polymer model.

Figure 5

We performed the same computation using the same parameters with our code to validate its accuracy. We got the comparable results as shown in figure 8.

1.7 Thesis scope

some scope stuff: Gaussian chain, Flory-Huggins interaction, pure Monte-Carlo way of finding binding sites.......

We also want to make this thesis a good starting point for students who are willing to continue this research topic or to apply the SCFT to other biological molecules. Therefore, we included an extensive explanation of the computational method, and we also raised questions for the unsolved problems.

The code we provide is easily understandable and accelerated by GPU with reasonably good speed......

2 Method SBS binding sites

Inspired by PRISMR [1] [2] [5] and a MeanFieldChromatin on GitHub [15], we developed our own algorithm to acquire bead color sequence from the contact map. In this section, we will introduce the workflow of this algorithm and discuss the reasons behind the steps.

The core idea behind this algorithm is to randomly change the color of a bead, and see if it makes the polymer behave more like the experimental input, by comparing the contact map. In principle, the contact map of the new bead sequence can be acquired from MD simulations for each iteration. However, due to the computational time it costs and the number of iterations needed, it is almost impossible and impractical to perform MD simulations for the Monte Carlo iterations. The method for constructing the contact map, known as the mean-field approximation (covered in Section 2.2), proposed by (reference !!!), is significantly faster without significantly sacrificing accuracy.

Note: some steps in this description are optional, and some values are not yet determined, because the input data might be different from case to case. And due to the scope and focus of this project, we didn't test a wide range of data to give the "perfect preset".

2.1 Preprocessing

The difference score (cost function) between the constructed contact matrix and experimental Hi-C map is computed at log scale. This ensures a point close to the diagonal line has a similar weight to a point off the diagonal line. Otherwise, the weight of the contact matrix is scaled by the power law of contact probability.

Hence, it is essential to convert the Hi-C map to a log scale at the first step and fill the points that contain NaN values with their adjacent values. It's optional to have the Hi-C map's value range clipped for better fitting in the subsequent steps.

2.2 Mean-field approximation: built contact matrix

The mean-field approximation predicts the contact matrix in a statistical way. The contact probability follows the power law measured from MD simulation of the coil or globule states of the polymer. If a pair of beads has the same color, their contact probability follows the globule state profile shown in figure 1b, and vice versa. In our code, a contour distance matrix is first calculated, and the contact matrix is calculated based on bead pairs and their corresponding contour distance.

The experimental Hi-C map's value can be anything between the coil and globule states' contact probability (and in some cases outside the range, but it is not considered in this case), rather than a "binary value" of one or another. To accommodate this issue, multiple beads were set for one pixel on the Hi-C map. When calculating the cost function, the constructed contact map, which has a higher resolution, is down-scaled to the same resolution as the Hi-C map with a Gaussian filter. This effectively averages the value of multiple pairs of beads to give a value that matches a point on the Hi-C map.

2.3 Matching power law profile

In practice, the Hi-C maps may have different resolutions. The contour distance matrix, which is used for calculating the contact matrix, is zoomed in this step, to rescale the contour distance per pixel (per bead). To get the accurate scale, a random bead sequence is used to generate a contact matrix, based on which a fall-off profile is calculated. The contour zoom factor and the intensity zoom factor are adjusted by comparing this profile with the Hi-C data's fall-off profile. These two profiles were matched to have the same power-law, and these two factors are used in the subsequent steps and remain unchanged.

2.4 Cost function

The cost function quantifies the quality of a given bead–polymer sequence. It is defined as the mean of the element-wise absolute difference between two matrices.

$$C(A,B);=;\frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n}\left|A_{ij}-B_{ij}\right|.$$

A and B are the constructed contact matrix and the Hi-C matrix. C is the cost function.

Additional terms can be added to this cost function to prevent overfitting. One of the simplest options is to add a term proportional to the total number of colors of the bead sequence. Other options evaluate the clustering behavior of beads with the same color.

2.5 Monte Carlo method and simulated annealing

The random bead sequence initialized at the beginning went through thousands of iterations of the Monte Carlo process to become the result. At each iteration, multiple trials were performed, each of which was assigned a new color to a randomly selected bead. The cost scores were calculated for each trial. The trial with the lowest cost score was chosen. This helps avoid the system from getting stuck in a local minimum.

The score of the best candidate sequence in the current trial, s_{new}, was compared with the score of the sequence from the previous iteration, s_{old}. The difference Δs was then used in a Metropolis acceptance step to decide whether to replace the previous sequence. Specifically, the new sequence was accepted with probability

$$A = \begin{cases} \exp(\frac{\Delta s}{T}), & \text{if } \Delta s > 0, \\ 1, & \text{otherwise.} \end{cases}$$

where T denotes the effective temperature, the algorithm starts with a relatively high temperature, and as the iterations go on, the temperature decreases. This helps the system find the global minima at the beginning, then has a more accurate fitting result at the end.

Figure 6 shows the bead color sequence inferred from the Hi-C map through this simulated annealing process. It also shows the constructed matrix, which has a high similarity compared to the Hi-C map input.

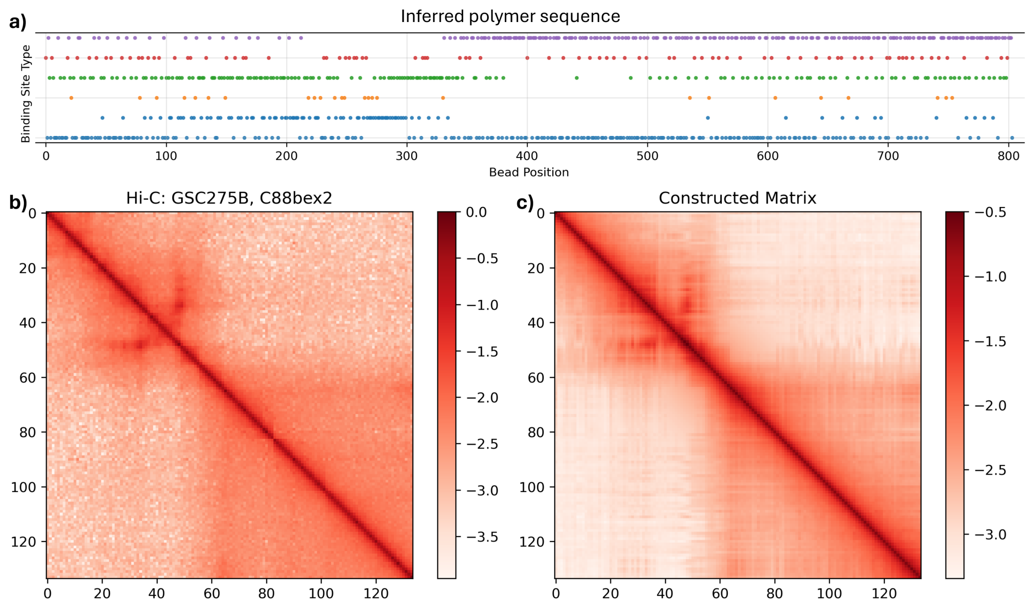
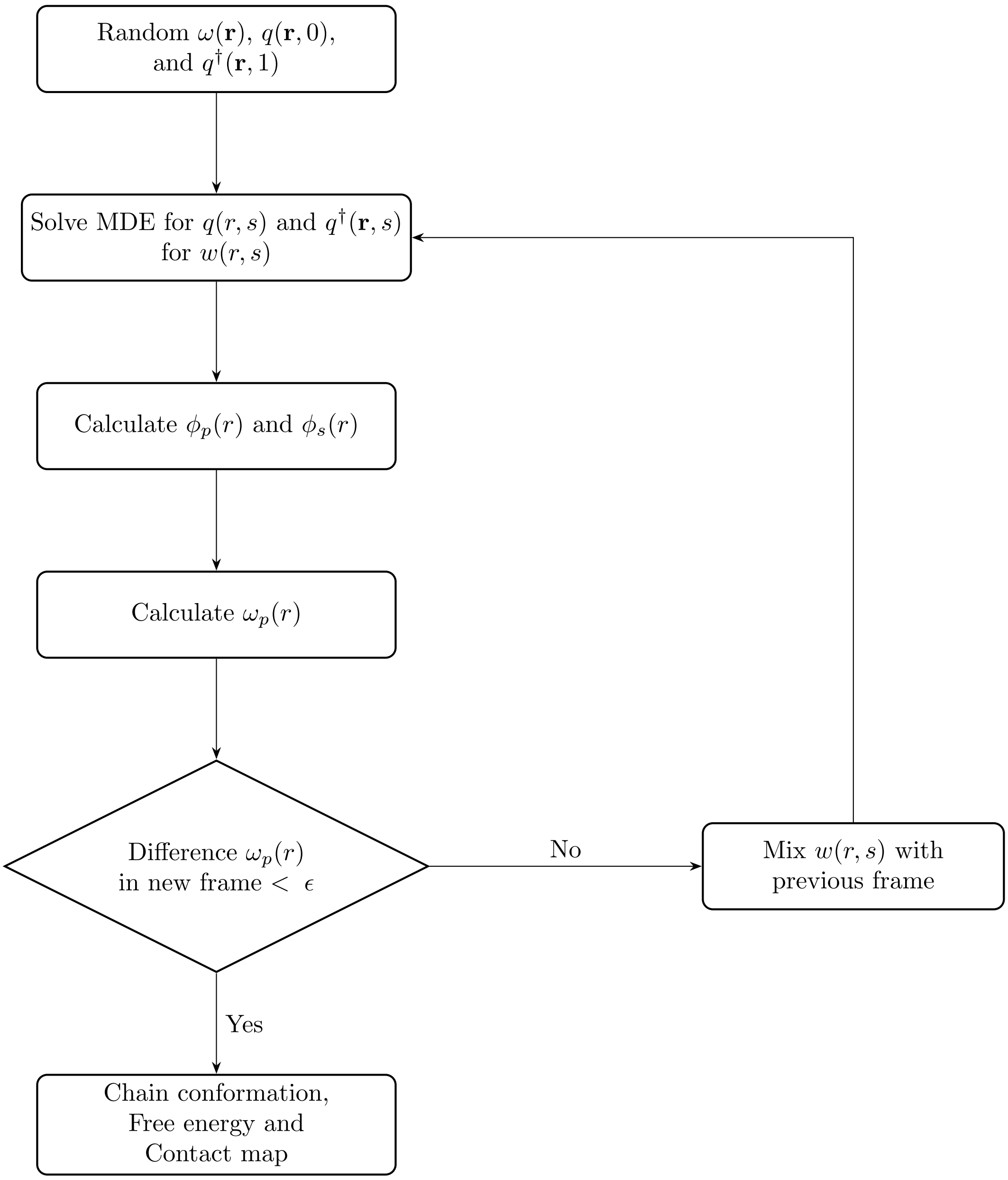


Figure 6: **(a)** The bead sequence inferred from the Hi-C map shown in **(b)**. **(c)** The contact matrix constructed from the inferred sequence.

3 Methods SCFT

Here, we introduce the SCFT approach we use in this study. The main idea of this mean-field approach is to iteratively compute polymer conformations and their conjugate chemical potential fields. In each iteration, the polymer conformation is calculated for a given field, then the field is updated based on the new polymer density. The field would eventually stay the same after enough iterations, if the computation converges, where the system is "self-consistent".



3.1 SCFT equations

still haven't shown how to derive the SCFT equations from the partition function

To illustrate the SCFT formulations, we consider an incompressible A+B homopolymer blend. The goal is to find the chemical potential fields ω_A and ω_B and the corresponding polymer conformation that minimizes the free energy of the polymer blend.

In the canonical ensemble, the free energy functional can be written as

$$F = \int d\mathbf{r} \left[-\omega_A(\mathbf{r})\phi_A(\mathbf{r}) - \omega_B(\mathbf{r})\phi_B(\mathbf{r}) + \frac{(\omega_B(\mathbf{r}) - \omega_B(\mathbf{r}))^2}{4\chi N} \right] - \bar{\phi}_A \ln Q[\omega_A] - \bar{\phi}_B \ln Q[\omega_B] \tag{1}$$

where ω is the chemical potential field, ϕ is the concentration field of polymer ranging from 0 to 1, $\bar{\phi}$ is the average density of polymer, χ is the interaction strength between A and B polymers, and Q is the single chain partition function.

In the continuous Gaussian model, the single chain partition function Q, subjected to a chemical potential field, is given by

$$Q(\omega) = \frac{1}{V} \int d\mathbf{r} q(\mathbf{r}, 1; \omega) \tag{2}$$

where the chain propagator $q(\mathbf{r}, s; \omega)$ satisfies the modified diffusion equation (MDE)

$$\frac{dq(\mathbf{r}, s; \omega)}{ds} = D\nabla^2 q(\mathbf{r}, s; \omega) - \omega_A(\mathbf{r})q(\mathbf{r}, s; \omega) \tag{3}$$

where s is the coordinate along the polymer contour, and D is the diffusion coefficient that equals to $\frac{b^2}{6}$, where b is the kuhn length. In equation 2, the contour length is normalized to 1. The propagator represents how the probability density of the polymer diffuses in the space as the contour goes. It is discussed in detail in section 3.3.

The concentration field of polymer $\phi(\mathbf{r})$ is calculated from integrating the product propagators that goes opposite directions along the contour

$$\phi_A(\mathbf{r}) = \int dsq(\mathbf{r}, s)q^\dagger(\mathbf{r}, s) \tag{4}$$

where $q^\dagger(\mathbf{r}, s)$ is the propagator in opposite direction. The probability density of the localization of each monomer is

$$\phi_p(\mathbf{r}, s) = q(\mathbf{r}, s)q^\dagger(\mathbf{r}, s) \tag{5}$$

Some literature use only one $q(\mathbf{r}, s; \omega)$, in this case

$$\phi_A(\mathbf{r}) = \int_0^1 dsq(\mathbf{r}, s)q(\mathbf{r}, 1 - s) \tag{6}$$

the $q(\mathbf{r}, s; \omega)$ flipped in s-direction.

The chemical potential field ω is given by the Flory-Huggins interaction:

$$\omega_A(\mathbf{r}) = \chi_{AB}\phi_B(\mathbf{r}) + \eta(\mathbf{r}) \tag{7}$$

The η in equation 7 encodes the incompressibility that is derived in the next section.

Flory's model assumes each particle is located on a uniform lattice and interacts with particles on the adjacent lattice. The Flory-Huggins χ parameter represents the additional energy when two different monomers get mixed. It is defined by

$$\chi_{AB} \equiv \frac{z}{2}\frac{1}{k_B T}(2u_{AB} - u_{AA} - u_{BB}) \tag{8}$$

where u is the interaction energy between a pair of monomers on adjacent lattice sites. z is the number of adjacent lattices.

3.2 SCFT equations for multi-block polymer with explicit solvent

The DNA polymer and proteins attached in combination occupy 10-30 percent (!!! reference) of the nucleus volume. The rest of the volumes are filled with water, ions, soluble proteins, and many other compounds and structures (!!! reference). We consider them as a homogeneous solvent made of colloid particles. The solvent and chromosome fill the space and are constrained by incompressibility:

$$\phi_s(\mathbf{r}) + \sum_p \phi_p(\mathbf{r}) = 1 \tag{9}$$

The chromosome is modeled as a polymer with many blocks, each block corresponding to a "color" assigned by the algorithm introduced in section 2. The chemical potential field of each type of polymer block follows

$$\omega_A(\mathbf{r}) = \sum_{A \neq B} \chi_{AB}\phi_B(\mathbf{r}) + \sum_B \chi_{BS}\phi_S + \eta((\mathbf{r}) \tag{10}$$

where the potential field for block P is based on the Flory-Huggins interaction with other blocks and the solvent. The chemical potential of the solvent follows a similar format:

$$\omega_s(\mathbf{r}) = \sum_B \chi_{BS}\phi_s(\mathbf{r}) + \eta(\mathbf{r}) \tag{11}$$

Note that the $\omega(\mathbf{r})$ for different blocks on the polymer may differ. Therefore, the field of the polymer is also a function of the contour $\omega_p(\mathbf{r}, s)$.

The solvent concentration field follows the Boltzmann distribution:

$$\phi_s = \frac{\bar{\phi}_s}{Q_s} e^{-\omega_s(\mathbf{r})} \tag{12}$$

By combining equation 12 and 9, the chemical potential of the solution can be expressed by the concentration of polymer blocks

$$\omega_s(\mathbf{r}) = -\ln\left(1 - \sum_p \phi_p\right) \tag{13}$$

which then can be substituted into equation 10 subtract equation 11 to give the expression for η, a Lagrange multiplier that encodes the incompressibility into the expression for the chemical potential field of the polymer blocks.

$$\eta(\mathbf{r}) = -\ln\left(1 - \sum_p \phi_p\right) - \sum_{A\neq B} \chi_{AB}\phi_B(\mathbf{r}) \tag{14}$$

The concentration of a polymer block type can be calculated by integrating the propagators on the contour sections of that block (color)

$$\phi_p(\mathbf{r}) = \int_0^1 ds q(\mathbf{r}, s) q^\dagger(\mathbf{r}, s) c(s) \delta(c, p) \tag{15}$$

where c_s describes the color of the polymer segments as a function of the contour position.

The free energy of the system is

$$\begin{aligned} F = &\underbrace{\rho_0 \int d\mathbf{r} \left[\sum_{A\neq B} \chi_{AB}(\mathbf{r})\phi_A(\mathbf{r})\phi_B(\mathbf{r}) + \sum_A \chi_{AS}\phi_A(\mathbf{r})\phi_s(\mathbf{r}) \right]}_{\text{interaction part}} \\ &\underbrace{-\rho_0 \int d\mathbf{r} \left[+ \sum_A \omega_A(\mathbf{r})\phi_A(\mathbf{r}) + \omega_s(\mathbf{r})\phi_s(\mathbf{r}) \right] - n_c \ln Q_c - n_s \ln Q_s}_{\text{entropic + constrain part}} \end{aligned} \tag{16}$$

where n_c is the number of copolymer chains in the system, n_s the number of solvent particles, and ρ_0^{-1} is the volume occupied by a monomer or particle. The first part of the expression corresponds to the interaction term of the free energy. The second part is the entropic and field constraint part. The free energy is invariant to a uniform chemical potential shift $\omega^*(\mathbf{r}) \to \omega(\mathbf{r}) + \omega_0$, the proof is given in the supplementary document [9]. We take ω_0 that satisfies $\sum_i \int d\mathbf{r} w_i(\mathbf{r}) = 0$. This helps to prevent the propagator $q(\mathbf{r}, s)$ from going beyond the floating point limit during numerical computation.

3.3 Modified diffusion equation

some derivations from the functional of the gaussian chain model

3.4 MDE solver

The equation 3 has the same shape as a heat conduction equation, where q is the temperature, s is time, and ω is the heat source/sink. There are two main types of methods for solving this equation: finite difference methods and Fourier methods. The finite difference Crank-Nicolson scheme can work with complicated boundary conditions, but slow for computation. In this research, we focused on the Fourier methods.

The Fourier transform in $\mathbf{r}$ space of the equation 3 is

$$\frac{\partial \hat{q}(\mathbf{k}, s)}{\partial s} = -D(2\pi)^2(k_x^2 + k_y^2 + k_z^2)\hat{q}(\mathbf{k}, s) - \widehat{\omega q(\mathbf{k}, s)} \tag{17}$$

where the wavenumber matrix $k_x^2 + k_y^2 + k_z^2$ corresponds to the frequencies in $\hat{q}(\mathbf{k}, s)$.

To calculate the evolution in the discretized s space, we implemented the semi-implicit method at first. The semi-implicit method has good accuracy and tolerance criteria. But it is not unconditionally convergent; sometimes, when Δs is too large, the result diverges.

$$\frac{q(\mathbf{r}, s + \Delta s) - q(\mathbf{r}, s)}{\Delta s} = D\nabla^2 q(\mathbf{r}, s + \Delta s) - \omega_p(\mathbf{r}, s) q(\mathbf{r}, s) \tag{18}$$

which in the Fourier space is

$$\hat{q}(\mathbf{k}, s + \Delta s) - \hat{q}(\mathbf{k}, s) = -\Delta s \left[D(2\pi)^2 (k_x^2 + k_y^2 + k_z^2) \hat{q}(\mathbf{k}, s) + \widehat{\omega q(\mathbf{k}, s)} \right] \tag{19}$$

And the $q(\mathbf{r}, s + \Delta s)$ can be written as:

$$\hat{q}(\mathbf{k}, s + \Delta s) = \frac{\hat{q}(\mathbf{k}, s) - \Delta s [\widehat{\omega q(\mathbf{k}, s)}]}{1 + \Delta s D (2\pi)^2 (k_x^2 + k_y^2 + k_z^2)} \tag{20}$$

The pseudo-spectrum method, or so-called split-operator method, introduced by Gilbert Strang, is an unconditionally converging method for the equation 3. At each forward step, the propagator is transformed by an operator $\mathcal{L}$

$$q(\mathbf{r}, s + \Delta s) = e^{\mathcal{L}\Delta s} q(\mathbf{r}, s) \tag{21}$$

where the operator $\mathcal{L}$ is composed of two parts: the diffusion operator $\mathcal{L}^D$ and potential field operator $\mathcal{L}^W$:

$$\begin{aligned} \mathcal{L} &= D\nabla^2 - \omega(\mathbf{r}) \\ &= \mathcal{L}^D + \mathcal{L}^W \end{aligned} \tag{22}$$

(maybe say something to make an analogy to quantum mechanics)

The second-order expansion in s of the $e^{\mathcal{L}\Delta s}$ equation 21 and is equal to

$$\begin{aligned} e^{\mathcal{L}\Delta s} &= 1 + (\mathcal{L}^D + \mathcal{L}^W)\Delta s + \frac{\Delta s^2}{2} [(\mathcal{L}^D)^2 + \mathcal{L}^D \mathcal{L}^W + \mathcal{L}^W \mathcal{L}^D + (\mathcal{L}^W)^2] + O(\Delta s^3) \\ &= e^{\mathcal{L}^W \Delta s/2} e^{\mathcal{L}^D \Delta s} e^{\mathcal{L}^W \Delta s/2} \end{aligned} \tag{23}$$

where in each step the $q(\mathbf{r}, s)$ is first multiplied by the half-step field operator $e^{\mathcal{L}^W \Delta s/2}$ in real space, then is transformed into the discrete Fourier space, where it is multiplied by the diffusion operator $e^{\mathcal{L}^D}$, then transformed back to real space, and lastly multiplied by $e^{\mathcal{L}^W \Delta s/2}$ again. When Δs is small, the $e^{\mathcal{L}^W \Delta s/2}$ effectively apply Euler method diffusion in the Fourier space and $e^{\mathcal{L}^D} = e^{2\pi D(k_x^2 + k_y^2 + k_z^2)\Delta s}$.

(I'm not sure this is 100% correct) The semi-implicit method has higher accuracy for Δs that satisfies the convergence criteria, compared to the pseudo-spectrum method with the same Δs.

3.5 Path integral

3.6 Computational problems

Here we address several things that need to be noticed when it comes to numerical simulation.

3.6.1 Contact map

In MD simulations, the contact map or contact probability can be calculated by sampling polymer segments within a cut-off distance from thousands of polymer conformations (phrase, reference !!!). However, in field theory, this can not be done due to the lack of exact localization of the chain. Alternatively, we define the contact map as the overlap of the probability of the polymer segments. The contact probability of two points i, j on the contour $P(i, j)$ is defined as

$$P(i, j) = \frac{1}{nx * ny} \sum_{x=0}^{nx} \sum_{y=0}^{ny} \phi_p(\mathbf{r}, i) * \phi_p(\mathbf{r}, j) \tag{24}$$

(!!! this equation needs to be improved to give a more comparable outcome)

3.6.2 Parameter normalization

It is conventional to normalize the contour length to 1 and scale other parameters accordingly. Here, we demonstrate how to normalize a chain with N_0 segments (contour length), each of length $b = 1$. We introduce the parameter N that has the same concept and value as N_{seg}, just for the convenience of the math expression. It's important to note that the interaction strength χN_0 and the morphology remain unchanged through this re-scaling.

$$\begin{aligned} N_{seg} = N_0 &\rightarrow N_{seg} = 1 \\ N = 1 &\rightarrow N = N_0 \\ b = 1 &\rightarrow b = \sqrt{N} \\ L = L_0 &\rightarrow L = L_0 \\ \text{eq.10} &\rightarrow \omega_A(\mathbf{r}) = \sum_{A \neq B} \chi_{AB} N \phi_B(\mathbf{r}) + \sum_{B} \chi_{BS} N \phi_S + \eta((\mathbf{r}) \\ \text{eq.12} &\rightarrow \phi_s = \frac{\bar{\phi}_s}{Q_s} e^{-\frac{\omega_s(\mathbf{r})}{N}} \end{aligned} \tag{25}$$

The Kuhn length b and box dimension L can be further rescaled by:

$$\begin{aligned} b = \sqrt{N} &\rightarrow b = 1 \\ L = L_0 &\rightarrow L = \frac{L_0}{\sqrt{N}} \end{aligned} \tag{26}$$

3.6.3 Boundary condition

The pseudo-spectrum method introduced in section 3.4 encodes periodic boundary conditions for the system. This is not the case for chromatin in the cell nucleus. Therefore, the no-flux boundary needed to be set for the simulations.

One easiest ways is to set the $\omega_p(\mathbf{r}, s)$ at the boundary to infinity. However, there would still be "leaks" across the boundary when the discretized contour length ds is large. The alternative way is to use the discrete cosine transformation (DCT), which provides a strictly no-flux boundary condition. We implemented the DCT via even expansion of the $\hat{q}(\mathbf{r}, s)$ in Fourier space and calculated its FFT. This gives the best compatibility with CuPy's FFT acceleration on GPU, despite the inefficiency due to the sine part being 0.

However, with this no-flux boundary condition, the polymer would crowd at the corners and sides of the box. This polymer conformation can avoid the mixing energy penalty with the solvent. In some research involving the detailed study of the interaction between the polymer and the boundary, an interaction potential function (e.g., electrostatic interaction with Debye screening) is added to the chemical potential field of the polymer. In this study, we don't look into the details of how chromatin interacts with the boundary, given the resolution and the length of the polymer (2M bp). We applied a trivial boundary condition that the field at the boundary equals $\chi_{PS}N$, the same value as if the boundary is pure solvent $\phi_{\text{boundary}=1}$. This effectively sets the boundary to not affect polymer conformation.

3.6.4 Mixing method

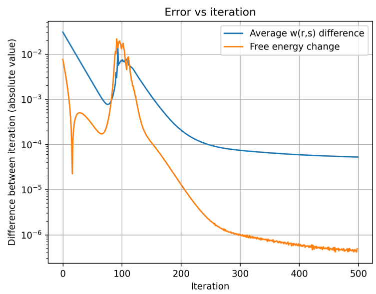
The simple mixing method is used to update $\omega_p(\mathbf{r}, s)$ each iteration

$$\omega_p^{i+1}(\mathbf{r}, s) = \theta \omega_p^{\text{updated}}(\mathbf{r}, s) + (1 - \theta)\omega_p^i(\mathbf{r}, s) \tag{27}$$

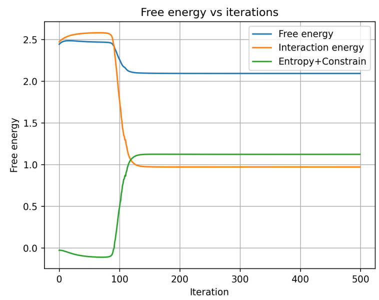
where θ is the mixing parameter of how aggressively the field is updated each iteration. In this study, θ is usually set to $\theta = 0.005$ for the chromatin, and $\theta = 0.05$ for the ABC tri-block polymer demonstration. Too large θ causes instability, and the result may not converge.

Figure 7 shows how the free energy changes with iterations for the ABC tri-block copolymer system we mentioned in section 1.6. The initial $\omega_p(\mathbf{r}, s)$ is a random matrix at the order of 10^{-3}. The free energy first slightly and slowly increased or decreased for dozens of iterations, then it decreased quickly, until it reached a stable value, which is the minimized free energy in SCFT !!!?. The polymer density in those iterations is shown in the figure 8. The morphology of the vesicle structure demonstrated in figure 5b develops mostly in the iterations where the free energy drops quickly.

It's important to stress that this morphology "development" only shows how the equilibrium state is calculated computationally. It does not indicate the time evolution of the system, since the SCFT method only describes a system at equilibrium.



(a) The free energy change and the average of difference between $\omega_p(\mathbf{r}, s)$ matrix each iteration. For the ABC triblock copolymer simulation of the figure 5a.



(b) The entropic and enthalpic part of the free energy in equation 16. The free energy decreases as the morphology of polymer develops.

Figure 7

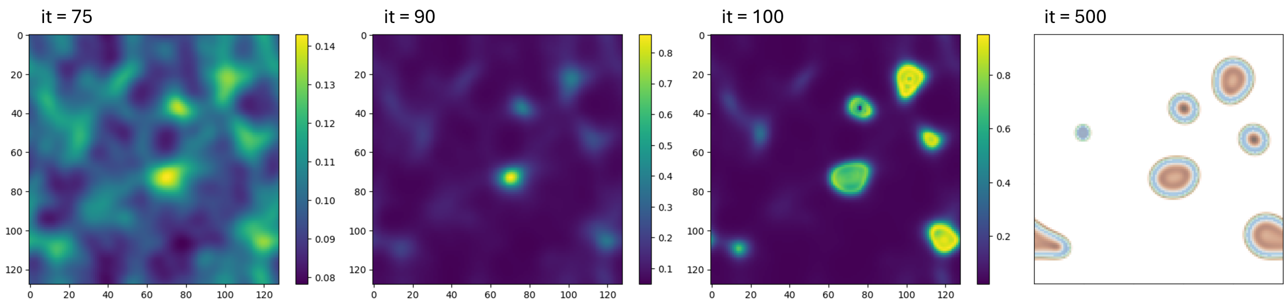


Figure 8: The polymer density changes with the iterations. During the 75th to 100th iterations, the morphology of self-assembled polymer vesicles developed. After 500 iterations, it shows a comparable result as figure5a.

We also tried to use the Anderson mixing method [16] (some math!!!) to accelerate the convergence. As indicated in the literature, the Anderson method is only used after some iterations of simple mixing when the morphology begins to develop. However, for the DNA polymer with a complicated block sequence, in our case, the most time-consuming part is this morphology development phase with simple mixing. Thus, the Anderson mixing method wasn't used.

3.6.5 Free energy

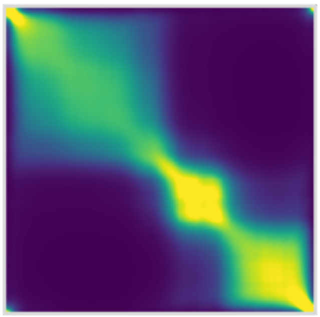
3.6.6 GPU acceleration

4 Results

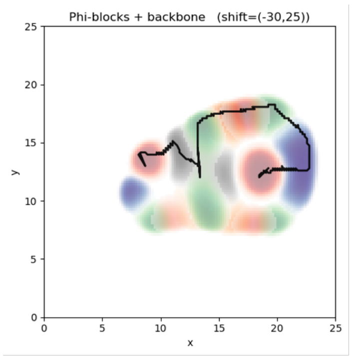
4.1 Parameter sweep

According to the Flory mean-field theory and SCFT literature, the χN parameter decides the phase behavior of the system......

4.2 Contact map



(a) The contact matrix (equation 3.6.1) of the polymer conformation computed in SCFT.



(b) The spatial distribution of polymer. The color clouds correspond to the concentration of different polymer blocks. The black line is the position where the possibility distribution of each polymer contour segment is the maximum.

Figure 9

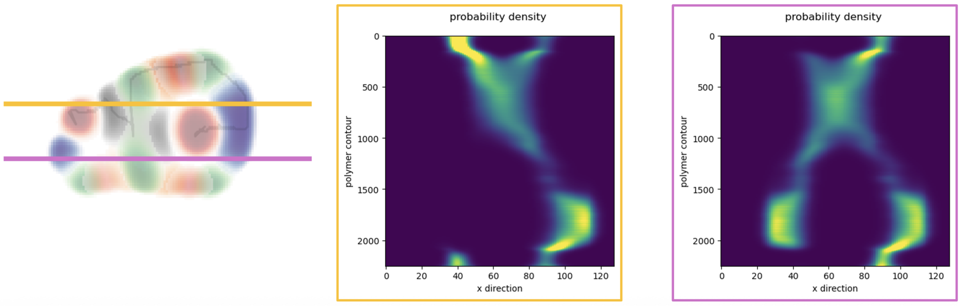


Figure 10: One-dimensional slices of the contour probability density. The positions of the two slices are indicated on the left.

4.3 Power law of distance

4.4 Density distribution to chain conformation

4.5 Radius of Gyration

a scheme showing the Rg and comparing with SBS results

4.6 Phase Behavior

4.7 Additional result: different boundary conditions' effect

5 Discussion

The limitations of the mean-field theory in biological systems......

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